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# The rich heterogeneous dial-a-ride problem with trip time prediction

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## Abstract

The dial-a-ride problem (DARP) involves designing vehicle routes to fulfill the door-to-door transportation requests of users where the goal is to minimize costs while satisfying transportation requests. In this paper, we introduce the rich heterogeneous DARP, which extends the generalized heterogeneous DARP to consider a fleet of buses and taxis, multiple depots, time windows at pickup and delivery locations, maximum ride and waiting times, and the possibility of an accompanying person. Our approach is based on a real service in Barcelona, and we also consider the variation in trip duration based on the time of day and day of the week. A predictive model is developed using machine learning techniques to estimate trip durations accurately. We apply our proposal to the daily door-to-door transportation of people with reduced mobility in Barcelona and demonstrate its superiority in terms of costs and quality of service by using the Gurobi optimizer. Additionally, we provide an analysis of the consequences of varying certain features on the costs and quality of service.

**Keywords:** dial-a-ride problem; heterogeneous vehicle; people transportation; vehicle routing

## 1. Introduction

The dial-a-ride problem (DARP) is a transportation optimization problem that consists of designing vehicle routes to fulfill the door-to-door transportation requests of users. Typically, the objective is to minimize costs while satisfying transportation requests. Each request requires picking up one or more passengers at a defined pickup point and then dropping off the passengers at the desired destination. The DARP is a variant of the pickup-and-delivery problem with time windows that incorporates constraints and objectives related to the quality of service provided to

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passengers, particularly those who are elderly or have disabilities. In recent decades, there has been extensive research on developing efficient models and solving the DARP problems.

In this paper, we present a new variant of the heterogeneous DARP (HDARP) that builds on the work of Parragh (2011) by adding additional features and by considering the rich nature of the problem. We refer to this problem as the rich HDARP (rHDARP), which is a variant of the HDARP that incorporates multiple constraints for tackling real-life scenarios, motivated by the daily door-to-door transportation of people with reduced mobility in the city of Barcelona. The rHDARP involves a heterogeneous fleet of vehicles and a group of users that are heterogeneous, utilizing buses and taxis, multiple depots, time windows at pickup and delivery locations, maximum ride and waiting times, and the possibility of an accompanying person member when needed. To solve the rHDARP, we propose a new mixed integer linear programming (MILP).

Our study is motivated by the Special Municipal Transportation Service (SMTS), which provides daily door-to-door transportation for people with reduced mobility in the city of Barcelona (Portell et al., 2022). This service is produced by the *Institut Municipal de Persones amb Discapacitat* (IMPD) and the *Àrea Metropolitana de Barcelona* (AMB), offering adapted buses and taxis for both regular and sporadic services. The sporadic services are for one-time or occasional requests, while the regular service includes a set of routes for daily trips scheduled yearly. Our focus in this paper is on finding solutions for the regular service, which is planned annually. The regular service caters to approximately 800 daily users and provides an accompanying person, a monitor, in the vehicle if needed. Our goal is to improve efficiency, to reduce costs, and to enhance the quality of service for the users of the regular routes by applying the rHDARP to the case of SMTS.

One key challenge in the rHDARP is the variation in trip duration based on the time of day and day of the week. We address this challenge by developing a predictive model that utilizes machine learning techniques to accurately estimate trip duration at various times of the day and on different days of the week. By considering the real-world constraints and objectives of the problem, our proposed solution provides a practical and efficient approach for solving the rHDARP. We also conduct an analysis of the time window and maximum ride time parameters to evaluate their impact on the cost and quality of the service in the SMTS case. By examining which parameters significantly affect various evaluation criteria, we provide guidelines to public transportation operators on how to design their dial-a-ride services. Previous research in this area includes the use of simulation to study time window settings by Quadrifoglio et al. (2008) and the examination of time windows and maximum ride time by Häll et al. (2015).

The main contributions of this paper are as follows:

- introduction of a new MILP to solve the rHDARP;
- introduction of a new objective function that considers the combination of buses and taxis, with different ways of calculating their costs, and the cost of an accompanying person if needed;
- case study of the real-case service SMTS in Barcelona, demonstrating the superiority of the solution in terms of costs and quality of service;
- development of a predictive model for estimating the trip duration using machine learning techniques; and
- analysis of the consequences in terms of costs and service quality of varying certain features such as the time windows and the maximum ride time of the user.

The remainder of the paper is organized as follows. In Section 2, we review the related literature on the DARP and the estimation of trip durations in DARP models. In Section 3, we formally define the problem and provide a general overview of our proposed solution. In Section 4, we describe the problem settings and present the mathematical model for solving the rHDARP with all relevant constraints. A predictive model is introduced in Section 5 for estimating the duration of trips. In Section 6, we present the computational experiments, including the definition of the real-life case SMTS, a description of the dataset, and the results of our proposed solution, including an analysis of the costs and quality of service. Finally, Section 7 contains concluding remarks, and we discuss future work.

## 2. Related literature

In this paper, we present research that builds upon two related areas of study: the DARP and trip duration predictive models. The DARP has long been a significant area of study in vehicle routing, and its associated literature encompasses a wide range of approaches and considerations, as surveyed by Cordeau and Laporte (2007). In the following section, we provide a comprehensive review of the relevant literature in these areas, highlighting the contributions of our work and how it extends upon previous research. Additionally, we explore the use of machine learning techniques for predicting and filling in missing data in vehicle routing problems, an area that has garnered increasing attention in recent years.

The DARP, as formally defined by Cordeau and Laporte (2003) and further elaborated upon by Cordeau (2006), involves the transportation of users who specify their pickup and delivery locations, and may also set time windows for their desired departure or arrival times. A fleet of vehicles is used to meet these transportation requests, with all vehicles based at a common depot. The objective of the problem is to minimize the costs of the vehicles while also fulfilling all requests within the constraints of the system, such as vehicle capacity, route duration, and maximum ride time.

The extensive literature on DARP, which ranges from exact to heuristic and metaheuristic methods, is evident in the work of Doerner and Salazar-González (2014). They offer an updated chronological overview of algorithms developed for the DARP, reflecting the dynamic exploration in the field.

The DARP further evolved to the heterogeneous case, HDARP, to accommodate the use of different types of vehicles and the varying requirements of users, particularly in the context of transporting elderly and disabled individuals who may require special accommodations such as a wheelchair or stretcher for transport and who may also require the presence of one or several accompanying persons (Liu et al., 2015). In our work, we build upon this concept by also considering the cost of the accompanying person. We assign an accompanying person to a vehicle when a user requests it. Braekers et al. (2014) further extended the HDARP to the multidepot case, allowing for the fleet to be allocated among multiple depots. In Parragh et al. (2012), the dynamics of driver constraints were explored, specifically introducing maximum route duration limits and mandatory lunch breaks. Additional research has been directed toward reducing the waiting time assigned to a vehicle while a user is aboard except for loading or unloading other users (e.g., Luo and Schonfeld, 2007; Schilde et al., 2014).

The utilization of taxi services for door-to-door transportation can present significant expenses, both in terms of financial cost and environmental impact. While buses may serve as a more cost-effective alternative, they lack the convenience and flexibility that taxis offer. Recent discussions on the evolving trends in urban passenger transportation optimization, noted by Filippi et al. (2023), emphasize the potential benefits of integrating different transportation methods. In our study, we explore a solution for the DARP that considers the trade-offs between using both taxis and buses, considering the advantages and limitations of each mode of transportation. Our approach is novel in comparison to previous studies, such as Vercraene et al. (2023), which considered the fixed distance and time-related costs of the vehicles, or Häll et al. (2015) and Tellez et al. (2018), who also added additional costs for waiting and excess ride time. To the best of our knowledge, no prior researchers have explored the cost implications of utilizing both taxis and buses in the DARP context.

The rich vehicle routing problem has been developed to address realistic scenarios by integrating multiple constraints that can provide a more accurate representation of real-world cases (Caceres-Cruz et al., 2014). In our case, we extend the DARP to formulate the rHDARP, which incorporates the additional constraints encountered in our problem domain.

Moreover, as we focus our study on real data, it is crucial to consider real-time variations in travel durations that may differ depending on the time and the day of the trip. Typically, in most published studies on the DARP constant travel times is assumed between two nodes, independent of the departure time. However, some researchers have used fixed speeds to calculate travel times (e.g., Braekers and Kovacs, 2016; Ouasaid and Saddoune, 2021), while others have attempted to enhance accuracy by employing average travel times that vary based on specific times of day (e.g., Eglese et al., 2006; Ehmke and Mattfeld, 2012).

In recent years, there has been increasing interest in utilizing machine learning algorithms for predicting trip durations. For instance, a study in New York showed various machine learning models for predicting travel times of taxi rides, and XGBoost emerged as the most accurate model (Manoharan et al., 2021). Similarly, in another study a deep neural network was proposed for predicting trip durations in Seattle by using traffic data collected by loop detectors on a freeway (Qu et al., 2019). The literature demonstrates that researchers have leveraged various data sources, such as traffic flow sensors, video image processors, GPS data, environmental data, smart card data, and social network data (Alsolami et al., 2020). A comprehensive survey of traffic prediction models is presented in Yuan and Li (2021), highlighting that machine learning based models can substantially enhance the accuracy of travel time prediction.

Table 1 compares our research on DARP with other relevant works in the field. By presenting the features we consider in our work and by highlighting which of those features are discussed in other papers, the table provides a comprehensive overview of the current state of the art. We include features related to both users and vehicles, such as time windows, maximum ride time, service time, waiting time, heterogeneous fleets, and accompanying persons. Additionally, we cover vehicle features, such as maximum ride time, heterogeneity of the vehicle, multiple depots, and different vehicle and staff costs. The table also features works that have calculated the duration of trips and evaluated the effect of various parameters. Overall, the table emphasizes the contributions of our work and provides a better understanding of where our research fits within the literature on the DARP.

Table 1  
Comparison of related literature on the DARP highlighting the contributions of our work

| Work                          | Users        |                |              |              |               | Vehicle             |           |               |            |                 | Parameter evaluation |
|-------------------------------|--------------|----------------|--------------|--------------|---------------|---------------------|-----------|---------------|------------|-----------------|----------------------|
|                               | Time windows | Max. ride time | Service time | Waiting time | Heterogeneous | Accompanying person | Max. time | Heterogeneous | Multidepot | Different costs |                      |
| Cordeau and Laporte (2003)    | X            | X              | X            | -            | -             | -                   | -         | -             | -          | -               | -                    |
| Parragh (2011)                | X            | X              | X            | X            | X             | X                   | -         | X             | -          | -               | -                    |
| Braekers et al. (2014)        | X            | X              | X            | -            | X             | -                   | X         | X             | X          | -               | -                    |
| Schilde et al. (2014)         | X            | X              | X            | X            | -             | -                   | X         | -             | -          | X               | -                    |
| Häll et al. (2015)            | X            | X              | X            | X            | X             | -                   | -         | X             | -          | X               | X                    |
| Liu et al. (2015)             | X            | X              | X            | -            | X             | X                   | X         | X             | -          | -               | -                    |
| Hu and Chang (2015)           | X            | X              | X            | X            | -             | -                   | -         | -             | -          | -               | -                    |
| Braekers and Kovacs (2016)    | X            | X              | X            | X            | X             | X                   | -         | X             | -          | -               | -                    |
| Tellez et al. (2018)          | X            | X              | X            | X            | X             | -                   | X         | X             | -          | -               | -                    |
| Gschwind and Drexl (2019)     | X            | X              | X            | X            | -             | -                   | X         | -             | -          | -               | -                    |
| Souza et al. (2020)           | X            | X              | X            | X            | X             | -                   | -         | X             | -          | -               | -                    |
| Vercraene et al. (2023)       | X            | X              | X            | X            | X             | -                   | -         | X             | -          | -               | -                    |
| Portell and Ramalhinho (2023) | X            | X              | X            | X            | X             | X                   | X         | X             | X          | X               | X                    |

### 3. Problem definition

In this section, we define the rHDARP problem we aim to solve in this paper. Our problem definition is based on a real-world problem that we describe in Section 6.1, where people with reduced mobility require door-to-door transportation services. We address a regular service, which is scheduled once per year. Our objective is to generate a transportation plan that fulfills all user requests while minimizing routing costs and considering the users' service quality. Our problem definition extends the HDARP problem originally proposed by Parragh (2011). To solve this problem, we must consider various factors affecting the overall transportation service quality and cost.

One crucial aspect to consider is the various user types, including those who require a regular seat or a wheelchair. The users' requirements are essential to consider when designing the transportation system, as they dictate the type of vehicles and their capacities needed. The vehicles used in the transportation system have different capacities and distributions of their space, including total capacities and capacities for wheelchairs or regular seats. Additionally, each vehicle has different operating costs, which can be either a fixed cost per trip or a cost per distance traveled. In our specific case, the transportation system consists of two types of vehicles: buses and taxis. Buses have a cost per kilometer, while taxis, adapted or nonadapted, have a fixed cost per user trip. These cost differences and capacity constraints must be considered when designing the optimal routes for the users, as they significantly impact the overall routing cost.

In addition to the transportation of users, the rHDARP problem may require the inclusion of an accompanying person, a monitor, in the vehicle to assist users with specific needs. It is important to note that a single monitor can assist multiple users in requesting a companion if needed. However, the accompanying person must be present in the vehicle if at least one user requires assistance. Including an accompanying person in the service incurs additional costs in the form of salary and the need for an extra seat in the vehicle. Therefore, when designing vehicle routes, it is essential to consider the number of users requiring assistance and the corresponding number of monitors needed to provide the service.

Another aspect that needs to be considered when solving the problem is the desired time window for pickup or delivery specified by the user. While users may request a specific pickup or delivery time, meeting their exact requirements may not always be efficient. To address this, we define a time window for the vehicle to pick up or drop off the user before or after their desired time. The time window allows the vehicle to arrive within a reasonable time frame to fulfill the user's request.

Limitations on the maximum ride time of each vehicle and passenger are also considered. These limitations guarantee that the vehicle cannot stay on the road for longer than a specified time, and each passenger cannot be on board for longer than a specific duration.

Waiting time is another constraint to minimize to provide an efficient and satisfactory transportation service. Waiting time is defined as the amount of time a user spends inside a vehicle while the vehicle is stationary, waiting to fulfill the time windows. This constraint ensures that users are kept from inside a vehicle for extended periods, which can lead to dissatisfaction and inconvenience. Waiting time can occur when a vehicle arrives to pick up a user before their designated pickup time window or when a vehicle arrives early to a delivery point before the designated delivery time window. By minimizing the waiting time of users while on board, we can provide a more efficient and satisfactory transportation service.

Furthermore, we must also consider the multidepot scenario, where each vehicle may have a different depot. This means that each vehicle may start and end at a different location. Furthermore, we must also consider passenger service time as part of the overall service duration. This means that the time it takes to load and unload each passenger must be accounted to determine the optimal route.

One important consideration for solving our real problem is the differentiation of trip duration at various times of the day. The duration of a trip can vary significantly depending on the amount of traffic at a given time. Unfortunately, we lack data on the duration of trips between specific points and times, which presents a challenge in optimizing our solution. To address this issue, we include a section in our research to create a predictive model that allows us to obtain the missing data.

The following is the formal description of the problem and the corresponding mathematical model.

#### 4. Problem settings and mathematical model

In this section, the MILP is formally described and modeled, based on the HDARP formulation by Parragh (2011).

##### 4.1. Problem setting and parameters

*Users and requests:*

- There are  $n$  transportation requests, each characterized by specific pickup and delivery locations and requested pickup or delivery times.
- The set of pickup locations for all requests is denoted as  $P = \{1, 2, \dots, n\}$ . A request  $i$  is associated with a unique pickup location  $i$ .
- The set of delivery locations corresponding to the pickup locations is denoted as  $D = \{n + 1, n + 2, \dots, 2n\}$ . For each request  $i \in P$ , the delivery location is  $i + n$ .
- After picking up a user from a location  $i \in P$ , the vehicle is not constrained to deliver them immediately. It can serve other requests, either further pickups or deliveries.
- Each request has a requested pickup time  $h_i$  for  $i \in P$  or delivery time  $h_{i+n}$ .
- There are  $f$  different types of users, represented by the set  $U = \{1, \dots, f\}$ . In this case,  $U = \{1, 2\}$ , where 1 represents a regular seat and 2 a wheelchair.
- The demand of users of type  $u \in U$  at pickup location  $i \in P$  is  $q_i^u$  for all  $u \in U$ . At the corresponding delivery location  $i + n$ , the demand is  $q_{i+n}^u = -q_i^u$ .
- For each pickup location  $i \in P$ , we define a binary parameter  $\gamma_i$  that indicates whether at least one user at that location requires an accompanying person. Specifically, if at least one user at pickup location  $i$  needs a companion,  $\gamma_i = 1$ ; otherwise  $\gamma_i = 0$ . Companions occupy a regular seat in the vehicle.

*Depots and vehicles:*

- There are  $m$  depots, each with a corresponding vehicle.

- The set of starting depots for the vehicles is denoted as  $O^+ = \{2n + 1, 2n + 2, \dots, 2n + m\}$ .
- The set of ending depots for the vehicles is denoted as  $O^- = \{2n + m + 1, 2n + m + 2, \dots, 2n + 2m\}$ . We allow vehicles to start in one depot and end in another. If all vehicles start and end in the same depot, only one of the sets is needed (e.g.,  $O = \{2n + 1, 2n + 2, \dots, 2n + m\}$ ).
- Set  $K$  is the set of  $m$  heterogeneous available vehicles. Each vehicle  $k \in K$  is associated with
  - a starting and an ending depot: we assume that the starting point for vehicle  $k$  is  $2n + k$  and the ending point is  $2n + m + k$ ;
  - each vehicle  $k \in K$  has a capacity  $Q_u^k$  for each type of user  $u \in U$ .

#### Graph:

- $G = (V, A)$  is a directed graph, where  $V$  is the set of all vertices and  $A$  is the set of all arcs.
- The set of all vertices is given by  $V = P \cup D \cup O^+ \cup O^-$ .
- The set of all arcs is given by  $A = \{(i, j) | i \in V \setminus O^-, j \in V - O^+, i \neq j\}$ .

#### Other parameters related to time and distance:

- $\delta$  is the size of the time window. We define a pickup time window  $[h_i - \delta, h_i + \delta]$  for each pickup location  $i \in P$  and a delivery time window  $[h_{i+n} - \delta, h_{i+n} + \delta]$ . Time windows are not defined for depots.
- $s_i$  is the service duration at pickup or delivery location  $i$  for all  $i \in P \cup D$ . No service duration at depots.
- $t_{ij}$  is the travel time from location  $i$  to location  $j$  for each  $(i, j) \in A$ .
- $d_{ij}$  is the distance traveled on arc  $(i, j) \in A$ .
- $T$  is the maximum ride time for each passenger.
- $\bar{T}$  is the maximum ride time for each vehicle, to not exceed the driver's working hours.
- $f^k$  is the fixed cost of vehicle  $k \in K$ .
- $c^k$  is the cost per unit distance for vehicle  $k \in K$ .
- $C$  is the cost of the companion per unit of time.
- $W$  is the maximum waiting time at a location.

#### 4.2. Decision variables

|                             |                                                                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| $x_{ij}^k \in \{0, 1\}$     | a binary variable that is 1 if vehicle $k$ visits and goes directly from location $i$ to location $j$ and 0 otherwise |
| $y_i^k \in \{0, 1\}$        | a binary variable that is 1 if vehicle $k \in K$ is assigned to location $i \in P \cup D$ and 0 otherwise             |
| $l_{iu}^k \geq 0$ , integer | the number of users of type $u \in U$ after visiting location $i \in V$ with vehicle $k \in K$                        |
| $w_i^k \geq 0$              | the arrival time of vehicle $k \in K$ at location $i \in V$                                                           |
| $z^k \in \{0, 1\}$          | a binary variable that is 1 if vehicle $k \in K$ needs a companion, and 0 otherwise                                   |
| $v^k \geq 0$ , integer      | the total time an accompanying person spends in vehicle $k \in K$                                                     |



### 4.3. Objective function

The objective of the optimization problem is to minimize the total cost, which consists of three parts:

1. Cost of vehicles with a cost per distance: cost of the total distance traveled by each vehicle, multiplied by the cost per distance unit for that vehicle.
2. Cost of vehicles with a fixed cost per user trip: cost associated with vehicles with a fixed cost per user trip, regardless of the distance traveled.
3. Cost of the accompanying person: cost associated with having a monitor in the vehicle.

The objective function can be written as:

$$\min \sum_{k \in K} c^k \sum_{(i,j) \in A} d_{ij} x_{ij}^k + \sum_{k \in K} f^k \sum_{i \in P} y_i^k + C \sum_{k \in K} v^k \quad (1)$$

is subject to the following constraints.

When constructing the origin–destination (OD) matrix with the distances and travel times between locations and depots, one must consider the type of vehicle being used. For taxis, which do not have a depot, a zero is written in the OD matrix for distances and travel times between locations and depots. For buses, the treatment of the depot depends on when the travel time begins: at the depot or when the first passenger is picked up.

### 4.4. Constraints

$$\sum_{j \in V} x_{ij}^k = \sum_{j \in V} x_{ji}^k = y_i^k \quad \forall i \in P \cup D, k \in K \quad (2)$$

$$\sum_{k \in K} y_i^k = 1 \quad \forall i \in P \quad (3)$$

$$\sum_{i \in P} x_{2n+k,i}^k - \sum_{i \in D} x_{i,2n+m+k}^k = 0 \quad \forall k \in K \quad (4)$$

$$w_j^k - w_i^k - t_{ij} - s_i + M_{ij}(1 - x_{ij}^k) \geq 0 \quad \forall (i, j) \in A, k \in K \quad (5)$$

$$w_j^k - w_i^k - t_{ij} - s_i - M_j(1 - x_{ij}^k) \leq W \quad \forall (i, j) \in A, k \in K \quad (6)$$

$$h_i - \delta \leq w_i^k \leq h_i + \delta \quad \forall i \in V, k \in K \quad (7)$$

$$w_i^k + s_i + t_{i,i+n} \leq w_{i+n}^k \quad \forall i \in P, k \in K \quad (8)$$

$$\begin{cases} l_{iu}^k \leq Q_u^k - z^k, & \text{if } u = 0, \\ l_{iu}^k \leq Q_u^k, & \text{if } u = 1. \end{cases} \quad \forall i \in P, k \in K \quad (9)$$

$$l_{ju}^k \geq l_{iu}^k + q_j^u - M_{iu}^k(1 - x_{ij}^k) \quad \forall (i, j) \in A, u \in U, k \in K \quad (10)$$

$$w_{i+n}^k - w_i^k - s_i \leq T \quad \forall i \in P, k \in K \quad (11)$$

$$w_{2n+m+k}^k - w_{2n+k}^k \leq \bar{T} \quad \forall k \in K \quad (12)$$

$$y_i^k \gamma_i \leq z^k \quad \forall i \in P, k \in K \quad (13)$$

$$w_{2n+m+k}^k - w_{2n+k}^k - \bar{T}(1 - z^k) \leq v^k \quad \forall k \in K \quad (14)$$

$$x_{ij}^k \in \{0, 1\} \quad \forall (i, j) \in A, k \in K \quad (15)$$

$$y_i^k \in \{0, 1\} \quad \forall i \in V, k \in K \quad (16)$$

$$z^k \in \{0, 1\} \quad \forall k \in K \quad (17)$$

$$l_{iu}^k \geq 0, \text{ integer} \quad \forall i \in V, u \in U, k \in K \quad (18)$$

$$v^k \geq 0, \text{ integer} \quad \forall k \in K \quad (19)$$

$$w_i^k \geq 0 \quad \forall i \in V, k \in K. \quad (20)$$

Constraints (2) ensure that every transportation is carried out by a single vehicle. Specifically, if a vehicle  $k$  enters a pickup or delivery location  $i$ , it must also leave that location and be assigned to it. Constraints (3) ensure that every request is satisfied exactly once. Constraints (4) ensure that any vehicle  $k$  leaving their starting depot must end at their final depot. If the vehicle does not require a depot, the distance and time to and from the depot are set to 0.

Constraints (5) and (6) linearize, using  $M_{ij}$ , the logical expression: if  $x_{ij}^k = 1 \Rightarrow 0 \leq w_j^k - w_i^k - t_{ij} - s_i \leq W$ . Constraints (5) set the arrival time at node  $j$  after the departure time from node  $i$ . The validity of these constraints is ensured by setting  $M_{ij} = (h_i + \delta) + s_i + t_{ij} - (h_j - \delta)$  and are based

on the work of Cordeau (2006). Constraints (6) impose a maximum waiting time  $W$  between nodes  $i$  and  $j$ , so that passengers do not wait too long between locations. The validity of these constraints is ensured by setting the big- $M$  value  $M_j = h_j + \delta - t_{ij} - s_i$ .

Constraints (7) enforce time windows at each location  $i \in V$ . Constraints (8) ensure that for each request  $i \in P$ , the pickup location is visited before the delivery location in the route.

Constraints (9) enforce the capacity constraint for each vehicle, where  $l_{iu}^k$  is the load carried by vehicle  $k$  on request  $i$  for user type  $u$ ,  $Q_u^k$  is the capacity of vehicle  $k$  for user type  $u$ , and  $z^k$  indicates whether vehicle  $k$  requires a companion. The constraints also ensure that if the companion is needed, the capacity of the regular seats of the vehicle is reduced by one (case  $u = 0$ ). Constraints (10) propagate the load along two consecutively visited points, considering each user type  $u \in U$  on each vehicle  $k$ . These constraints linearize the nonlinear constraints  $l_{ju}^k \geq (l_{iu}^k + q_j^u)x_{ij}^k$  by setting  $M_{ju}^k \geq \min\{Q_u^k, Q_u^k + q_j\}$ .

Constraints (11) limit the total ride time for all requests  $i \in P$  (including service time and waiting time). Constraints (12) ensure that the duration of each route is less than the maximum allowed time  $\bar{T}$ .

Constraints (13) ensure that vehicle  $k \in K$  requires a companion if user  $i$  needs a companion and is picked up by vehicle  $k$ . Constraints (14) enforce the time traveled by a companion on vehicle  $k$  if there is one. These constraints linearize the nonlinear constraint  $(w_{2n+m+k}^k - w_{2n+k}^k)z^k \leq v^k$ . The validity of these constraints is ensured by setting the big- $M$  value as  $\bar{T}$ . Finally, constraints (15)–(20) define the decision variables.

To simplify the problem and make it more tractable, we have made some assumptions about the structure of our model. Specifically, we have assumed that all taxis and buses operate between a depot and various other locations. However, this assumption does not hold for taxis or buses that begin their routes once they pick up their first passenger and end them when they drop off their last passenger. To account for these cases, we set the distance and duration between the depot and all other locations, as well as between any other locations and the depot, to zero. This approach allows us to introduce this assumption without introducing any limitations or biases into the model.

To address the challenge of obtaining data on the duration between locations, we developed a method for estimating duration, presented in the following section. An OD matrix with duration was necessary to solve the model, and our approach enabled us to overcome the lack of these data. To account for variations in travel time due to the time of day and day of the week, we generated an OD matrix by calculating the duration between pickup points while considering the desired pickup time specified by the user. Specifically, for each pickup point  $i$ , we calculated the duration between  $i$  and any other point assuming a departure time of the user-specified pickup time (e.g., 5 p.m. on a Wednesday). This approach allows us to obtain more accurate travel time estimates and to optimize transportation routes accordingly.

In this study, we use an exact solution approach and a state-of-the-art commercial solver (Gurobi) to solve the problem for small instances. Other approaches in the literature use meta-heuristic techniques to solve larger instances (Ho et al., 2018). Our goal in using an exact solution approach is to demonstrate the improvements made by our model and to lay the groundwork for using this approach to solve larger instances in the future.

Table 2  
OD duration matrix for different time categories

|            | Depot      | Destination 1 | Destination 2 | Destination 3 | ... | Destination $n$ |
|------------|------------|---------------|---------------|---------------|-----|-----------------|
| Depot      | 0          | $d_{01}^p$    | $d_{02}^p$    | $d_{03}^p$    | ... | $d_{0n}^p$      |
| Origin 1   | $d_{10}^p$ | 0             | $d_{12}^p$    | $d_{13}^p$    | ... | $d_{1n}^p$      |
| Origin 2   | $d_{20}^p$ | $d_{21}^p$    | 0             | $d_{23}^p$    | ... | $d_{2n}^p$      |
| Origin 3   | $d_{30}^p$ | $d_{31}^p$    | $d_{32}^p$    | 0             | ... | $d_{3n}^p$      |
| ...        | ...        | ...           | ...           | ...           | 0   | ...             |
| Origin $n$ | $d_{n0}^p$ | $d_{n1}^p$    | $d_{n2}^p$    | $d_{n3}^p$    | ... | 0               |

The table shows the duration (in minutes) of the trip between points  $i$  and  $j$  (rows and columns) in the time category  $p$ .

## 5. Predictive modeling for trip duration estimation

The methodology for obtaining the OD duration matrix at a given moment in time  $p$  is described in this section. The matrix (shown in Table 2) has row indices representing the origin locations and column indices representing the destination locations. The value at position  $ij$  represents the duration of a trip from origin  $i$  to destination  $j$  at a given moment  $p$ .

The creation of a duration estimator is important because it allows us to overcome the difficulties related to time, the number of query limitations, and the cost associated with creating OD matrices each time a different problem has to be solved. Public organizations and many others may need more resources to use applications that have a cost each time. They are free applications that can provide duration estimates, but they often do not allow users to specify the time of day or week or may have limitations on the number of queries that can be made. In this study, we present a solution to estimate trip durations without relying on external applications. This solution applies to a wide range of scenarios and can be used to support decision-making in transportation planning.

To create these OD matrices, we must be able to predict the duration of a trip from a given origin, destination, and date. This allows us to predict the value of each  $ij$  position in the matrix. To do this, we use predictive modeling techniques to train a model on a dataset of known trip durations between two points and use the trained model to make predictions on new data. Next, considering the day and hour of the day, we apply a “traffic” factor obtained by clustering techniques. This factor accounts for variations in travel time due to differences in traffic conditions during the different hours and days of the week.

We use two different techniques to predict the duration of trips. The first technique is based on the correlation between trip distance and duration. However, this method has lower accuracy because distance does not consider factors such as traffic lights or the time and day of the week when the trip is taken. Despite its lower accuracy, this method can be used as an approximation without needing a machine learning model or large amounts of data. To calculate the distances between points, we use the Manhattan distance. The Manhattan distance between two points is defined as the distance between the two points measured along axes at right angles, that is, given two distinct points  $P = (p_x, p_y)$  and  $Q = (q_x, q_y)$ , the Manhattan distance is  $d(P, Q) = |p_x - q_x| + |p_y - q_y|$ .

The second technique uses machine learning techniques to obtain more accurate predictions. In this case, we consider additional information such as the time and day of the trip and specific regional information (such as the neighborhoods of the origin and destination). We do not consider

other real-time factors, such as weather or road construction, nor do we consider the day of the year or month. This decision was made because we offer a solution intended for annual scheduling.

To implement the second technique, we divide the problem into two distinct steps: predictive modeling and traffic clustering. In the predictive modeling step, we explore and train a predictive model (see Section 5.1.2) to estimate the duration of the trip given the origin and destination points and other information. In the traffic clustering step, we examine the differences in trip durations across various days of the week and hours of the day and incorporate these variations into the model's predictions by adding the appropriate traffic duration. By combining these two steps, we can more accurately forecast trip durations and generate OD matrices that account for variations in travel time due to the time of day and day of the week.

### 5.1. Predictive modeling: training a model to forecast trip durations

#### 5.1.1. Data preparation

To create the training dataset for our predictive model, we collect data that can be used to train and evaluate the model. Each dataset row consists of the origin, destination, and actual duration provided by our data source OpenStreetMap (OpenStreetMap contributors, 2017). In addition, we include the neighborhood of origin and destination (other types of aggregations, such as districts or zones, could work as well) and the Manhattan and Haversine distance between the points. These additional features were shown to improve the performance of the model in our experiments. Training data can be expanded or reduced as desired, improving model accuracy or simplifying input data and model training time.

Since some evaluated models can only process numeric vectors, we convert the categorical feature attributes (such as the origin and destination neighborhoods) into numerical values by using the one-hot encoding method. This allows us to use a broader range of models and to achieve better prediction performance.

#### 5.1.2. Predictive modeling

In this section, we describe the steps involved in creating a predictive model to estimate the travel time between two locations. Our approach is outlined in Fig. 1.

First, we use the data preparation methodology described in Section 5.1.1 to create a training dataset. This dataset is then split into training and test sets, with 80% of the data used for training and the remaining 20% used for testing.

We evaluate four different regression algorithms for predicting trip duration: XGBoost Regressor (Chen and Guestrin, 2016), Random Forest (RF) Regressor, Support Vector Machine (SVM) Regressor, and Linear Regression. These models were selected based on their performance in similar scenarios in the literature (Manoharan et al., 2021).

Next, we use grid search to optimize the hyperparameters of the regression models and  $k$ -fold cross-validation to evaluate and improve the reliability of the models. In our experiments, we use fivefold cross-validation.

Finally, we compare the performance of the selected models using multiple evaluation criteria: root mean squared error (RMSE),  $R$ -squared ( $R^2$ ), and mean absolute error (MAE). The model with the best performance is selected as the final predictive model for estimating trip duration.

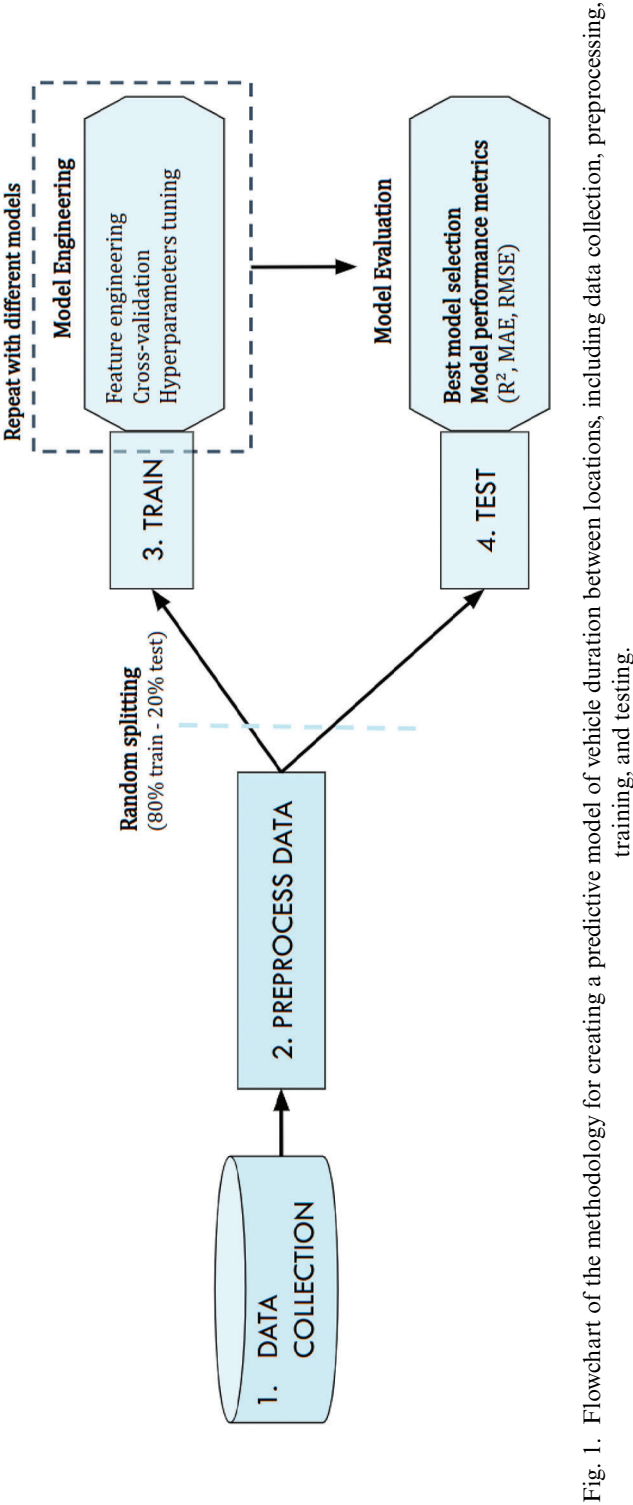


Fig. 1. Flowchart of the methodology for creating a predictive model of vehicle duration between locations, including data collection, preprocessing, training, and testing.

By using the predictive model, we are able to generate a preliminary OD matrix that provides predicted durations for each OD pair. Each cell in this matrix represents the predicted duration of a trip from the origin (represented by the row) to the destination (represented by the column). In the next step, we incorporate the traffic factor into these predictions, adjusting the values based on the day of the week and the hour of the day. The calculation of the traffic factor is discussed in the following section.

### 5.2. *Traffic clustering: incorporating variations in travel time due to time of day and day of the week into predictive modeling*

In this section, we describe a methodology described in Fig. 2 for clustering weekdays and hours based on their similar trip duration behavior and computing the corresponding multiplication factor for each cluster. To achieve this, we require a dataset containing various OD pairs along with their actual trip duration for each combination of weekdays and hours (as shown in Table 6). It is crucial to have samples for every possible time combination to study their behavior accurately.

To reduce the dimensionality of the data, we apply principal component analysis (PCA) and use the elbow method to determine the appropriate number of clusters for the  $K$ -means algorithm. We then apply  $K$ -means to obtain the clusters. We also performed this process without using PCA, and we obtained the same results.

We cluster the hours and weekdays into  $K$  different clusters, and we assume that the travel time between two points is the same for different times if they belong to the same time cluster. The specific number of clusters and their behavior can vary depending on the region being examined.

After obtaining the  $K$  different clusters, we calculate the multiplication factor for each cluster. First, we calculate the mean duration time for each OD pair within each cluster. We then determine the multiplication factor for each OD point by dividing the duration time of the OD points of the cluster by the minimum duration of the OD points across all clusters.

Once we have calculated the multiplication factor for each cluster for each OD point, we obtain the final factor for each cluster by taking the mean of the multiplication factors across all the OD points.

In this way, given the predicted duration of a new OD pair, we determine which cluster it belongs to and multiply the duration prediction by the corresponding cluster factor, considering the extra hour-weekday factor. This approach enables us to estimate travel time more accurately and to optimize transportation routes accordingly.

Finally, we use this matrix as input for solving the rHDARP optimization problem.

## 6. Computational experiments

In this section, we present the computational experiments conducted to assess and compare the quality of our approach. The experimental study is carried out with a case study of the city of Barcelona.

All programs are implemented in Python 3.8. The solutions to the routing problem are obtained with an exact approach. The applied solver is Gurobi 9.5.1. (Gurobi Optimization, LLC, 2023). Default settings are applied. The experiments are carried out on an HP ProBook 440 G7



Fig. 2. Schematic of the methodology for clustering days of the week and hours by similar behaviors.



computer with an Intel Core i7-10510U CPU, 1.80 GHz, and 16 GB RAM memory. The data preprocessing and the prediction are implemented by using Python 3.8 and Python third-party libraries: numpy, pandas (The Pandas Development Team, 2020), geopandas, and scikit-learn (Pedregosa et al., 2011).

In the following, we present the case study of the city of Barcelona in Section 6.1 and introduce the datasets used in Section 6.2. In Section 6.3, we examine the performance of our predictive model for obtaining the trip durations needed as input data for solving the rHDARP problem. In Section 6.4, we solve the routes and compare the results obtained with the current routes in use. Additionally, we examine the results and analyze the trade-off between routing cost and different time windows.

### 6.1. Case study definition

The case study presented in this paper focuses on the SMTS in Barcelona, a door-to-door public transportation service specifically designed for people with reduced mobility. The SMTS is provided by the IMPD in collaboration with the AMB and operates throughout the year. The service is available for both sporadic and regular trips, with the latter referring to routes that users regularly make over time. We are facing the problem of regular service.

Regular trips are those that users plan to make over time. Users can request trips in July to be carried out throughout the year. Users must provide information about their desired pickup time, origin, and destination when requesting a trip. They must also indicate if they are in a wheelchair and whether they require an accompanying person to be in the vehicle during their journey. The extra cost associated with having an accompanying person in the vehicle must be considered.

Routes are then planned according to the user's requests, considering the vehicle capacity, time windows, and the maximum allowable time for a ride. Since the SMTS operates daily throughout the year, trip durations are estimated rather than determined in real time.

The SMTS fleet comprises buses of varying capacities and adapted and nonadapted taxis. Adapted taxis are equipped to accommodate individuals using wheelchairs. The cost of bus travel is calculated based on the distance traveled from the first passenger picked up until the last passenger dropped off. In contrast, the cost of a taxi ride is fixed per trip.

It is important to note that the SMTS calculates bus travel cost considering the distance from the first to the last passenger, rather than to or from the depot. Consequently, in the duration and distance OD matrix, distances and durations between the depot and any other point, as well as between any point and the depot, are assigned a value of 0.

To improve the current system, we compare the proposed improvements to the current service based on data obtained from user requests and the routes implemented by the SMTS. Specifically, in Section 6.2, we provide more detailed information about the distribution of the vehicles and user requests.

### 6.2. Dataset

The datasets used in this study consist of three types of data: user, fleet, and trip duration. The user data contain information about the requests made by users, including their routes and preferences. The fleet data contain information about the vehicles and routes of the fleet. The trip-duration data

Table 3  
Summary of users in different routes

| Route | No. of users | No. of users with wheelchair | No. of users with companion |
|-------|--------------|------------------------------|-----------------------------|
| A     | 7            | 3                            | 1                           |
| B     | 8            | 6                            | 2                           |
| C     | 6            | 4                            | 1                           |
| D     | 5            | 3                            | 4                           |
| E     | 7            | 2                            | 2                           |

The table shows the total number of users (column No. of users) and the number of users with wheelchairs or that need a companion (columns No. of users with wheelchair and No. of users with companion) for each route (A, B, C, D, and E).

are used to create our predictive model to use it as extra information for solving our problem. The structure of the various datasets shown is described in more detail below:

- *User data:* In the user data, we have information about the requests made by users, including their origin and destination locations, desired pickup time, and any special needs (such as a wheelchair or a companion person). We also have information about which route they have been assigned in the current system and when they are being picked up. This allows us to compare our created route with the one currently in use. SMTS satisfies a daily demand in regular service of approximately 800 users. In this case, we are treating 33 users, which can be expanded with other techniques in the future. The IMPD provides this information.

Table 3 summarizes the five routes selected for our experiments. It shows the number of users in each route, the number of users with a wheelchair, and the number of users who need a companion. For privacy reasons, we are not offering the data of the users' requests or the routes being assigned by the SMTS.

- *Fleet data:* The fleet data include information about the vehicles available for the SMTS and their capacities. We have a large fleet of buses and taxis, but we use only a small number in our experiments. There are 24 minibuses, each with different capacities and distributions (as shown in Table 4). There are adapted and nonadapted taxis. We consider that the maximum capacity of an adapted taxi is one regular seat and one for a wheelchair, and that of a nonadapted taxi is two regular seats.
- *Trip-duration data:* We created two trip-duration datasets to obtain our predictive model. The first dataset is used to train and test the predictive model for estimating the duration between locations (see Table 5). This dataset has been created by selecting 25,000 OD trips in the city of Barcelona, adding the origin and destination neighborhood of the points and the Manhattan distance between the points, and finally obtaining the duration of going from one point to another. This duration was obtained by using OpenStreetMap data (OpenStreetMap contributors, 2017) obtained from the Open Source Routing Machine (OSRM) (Luxen and Vetter, 2011). The use of this dataset is discussed in detail in Section 6.3.

The second dataset, the traffic clustering dataset, is used in this study to cluster the hours and days of the week, as described in Section 5.2. The dataset has been created by processing data from Open Data BCN (Open Data BCN—Ajuntament de Barcelona, n.d.), which provides in-

Table 4  
Capacity of buses with different numbers of seats and wheelchair spaces

| Vehicle id. | No. of seats | No. of wheelchair | No. of vehicles |
|-------------|--------------|-------------------|-----------------|
| 1           | 2            | 4                 | 1               |
| 2           | 2            | 4                 | 1               |
| 3           | 6            | 1                 | 1               |
| 4           | 3            | 6                 | 1               |
| 5           | 4            | 6                 | 1               |
| 6           | 5            | 6                 | 10              |
| 7           | 7            | 5                 | 4               |
| 8           | 10           | 4                 | 5               |

The table shows the number of seats (column No. of seats), the number of wheelchair spaces (column No. of wheelchair), and the total number of vehicles (column No. of vehicles) for each combination of seats and wheelchair spaces.

Table 5

The table presents data on trips between various origin and destination pairs in the city, including their coordinates, neighborhood names, the distance between origin and destination, and time of travel

| Origin coordinates |           | Origin neighborhood | Destination coordinates |           | Destination neighborhood | Manhattan distance (km) | Real duration (s) |
|--------------------|-----------|---------------------|-------------------------|-----------|--------------------------|-------------------------|-------------------|
| Latitude           | Longitude |                     | Latitude                | Longitude |                          |                         |                   |
| 41.346176          | 2.168365  | La Marina de Port   | 41.146176               | 2.202012  | El Poble Sec             | 1.3                     | 245               |
| 41.129183          | 2.153231  | El Clot             | 41.002013               | 1.987987  | Hostafrances             | 8.3                     | 623               |
| 41.488787          | 2.093212  | La Barceloneta      | 41.321221               | 2.038918  | El Clot                  | 5.1                     | 548               |

Table 6

This table presents the traffic clustering dataset, with the duration of trips in seconds between various origin–destination pairs grouped by hour and weekday

|                     | Origin–destination 1 | Origin–destination 2 | Origin–destination 3 | Origin–destination 4 | ... | Origin–destination 78 |
|---------------------|----------------------|----------------------|----------------------|----------------------|-----|-----------------------|
| Weekday 1 - Hour 1  | 365                  | 984                  | 1200                 | 231                  | ... | 656                   |
| Weekday 1 - Hour 2  | 365                  | 972                  | 1190                 | 192                  | ... | 675                   |
| Weekday 7 - Hour 23 | 362                  | 960                  | 1150                 | 201                  | ... | 648                   |
| Weekday 7 - Hour 24 | 361                  | 972                  | 1125                 | 203                  | ... | 693                   |

formation on 78 sections within the city of Barcelona that were collected through sensors. The data include the travel duration from the start to the end of each section every 15 minutes.

To create the traffic clustering dataset, we compute the average travel time for each section during every hour of the week, allowing us to observe the differences in travel duration during various hours of the day and days of the week. The resulting dataset, represented in Table 6, comprises  $7 \times 24$  rows, representing each day of the week with their 24 hours, and 78 columns representing

Table 7

The table presents the results of applying the predictive models

|                   | $R^2$       | MAE          | RMSE         |
|-------------------|-------------|--------------|--------------|
| Linear regression | 0.87        | 77.34        | 101.52       |
| SVM               | 0.85        | 82.41        | 108.36       |
| RF                | 0.93        | 55.58        | 73.02        |
| XGBoost           | <b>0.98</b> | <b>28.12</b> | <b>37.51</b> |

The models were evaluated using three metrics:  $R$ -squared, mean absolute error, and root mean squared error. The highest  $R$ -squared and lowest error rates for each metric are denoted in bold.

different sections within the city. Each cell within the dataset contains the duration of traveling the corresponding section during the given day and hour.

We investigate the correlation between trip duration and day hours by using the traffic clustering dataset. Further details on the application of the dataset are discussed in Section 6.3.1.

### 6.3. Predictive model evaluation for data preparation

To solve the routing problem, we have seen that we need to create the OD duration matrix. In this section, we present the results of our predictive model, which was obtained by using the methodology described in Section 5.

#### 6.3.1. Predictive model

In this section, we show the results of training several predictive models on the data presented in Table 5.

Table 7 presents the performance of the four models on the test dataset. The models are linear regression, SVM, RF, and XGBoost. The table shows the value of three evaluation metrics for each model:  $R^2$ , MAE, and RMSE.

$R^2$ , or the coefficient of determination, measures how well a model fits the data. It ranges from 0 to 1, with a higher value indicating a better fit. MAE measures the average magnitude of the errors in a model's predictions. It is calculated as the sum of the absolute differences between the predicted and actual values divided by the number of predictions. RMSE measures the average magnitude of the errors in a model's predictions. It is calculated as the square root of the average squared differences between the predicted and actual values.

Based on the values in the table, we can see that the XGBoost model performs the best, with an  $R^2$  value of 0.98, an MAE of 28.12 seconds, and an RMSE of 37.51 seconds. This indicates that the model is able to make highly accurate predictions on the given dataset.

After tuning the hyperparameters of XGBoost, the final parameters are determined to be *col-sample\_bytree*: 1.0, *eta*: 0.3, *gamma*: 0.3, *learning\_rate*: 0.1, *max\_depth*: 9, *min\_child\_weight*: 7, *n\_estimators*: 1000, *subsample*: 0.7. These values are found to provide the best performance on the given dataset.

Finally, the XGBoost regressor is selected as the final predictive model. We have seen that we have achieved a high accuracy level in estimating the vehicle route duration between points.

Furthermore, these results are used for the creation of annual routes, and therefore a higher precision is not necessary. Overall, the XGBoost regressor has proven to be a reliable and effective model for predicting vehicle travel times in our transportation network.

### 6.3.2. Results of clustering weekdays and hours with similar trip duration behaviors

By using the dataset presented in Table 6, we want to cluster the weekdays and hours with similar behavior in terms of trip duration as done with the methodology in Fig. 2. Since there are many columns for the number of existing rows, we apply PCA. We reduce the dataset to five principal components, maintaining a variance of more than 95% (96.14%).

We apply the elbow method to obtain the optimal number of clusters from 2 to 10, obtaining the within-cluster sum of squares (WCSS) for each  $K$ -value. WCSS is the sum of squares of the distances of each data point in all clusters to their respective centroids. We select the value for  $K$  based on the rate of decrease in WCSS. The number of clusters obtained is 4.

Figure 3 displays the clustering results, with the rows indicating the days of the week and the columns representing the 24 hours of the day. Each cell in the table is color-coded to represent one of the four clusters obtained, reflecting the relative traffic levels. Light yellow represents times with the shortest trip duration and red represents times with the longest trip duration.

The clustering results provide valuable insights into the traffic patterns of Barcelona. For instance, we can observe that the highest traffic volumes occur during the morning and evening rush hours, with Fridays standing out as a particular case due to the shorter workday in many jobs, resulting in an earlier peak in traffic than that on other days of the week. Additionally, the prevalence of light yellow cells on weekends indicates low traffic volume during these days.

In summary, Fig. 3 presents a clear visual representation of the traffic patterns in the city and reaffirms the effectiveness of our clustering methodology in identifying traffic patterns.

## 6.4. Solving the DARP

In this section, we evaluate the effectiveness of using our rHDARP model to improve the current transportation scheme of the SMTS. We do this by comparing the results of the current SMTS routes with our optimized solutions for the routes presented in Section 6.2. We use Gurobi to solve the model described in Section 4.

First, we begin by comparing the performance of the current SMTS routes with our optimized solutions. Second, we analyze the impact of varying the time windows on the results of the routes. This allows us to determine if using the rHDARP model can lead to significantly lower costs and shorter user times than those of the current system.

### 6.4.1. Parameter settings

In our transportation model, we set the following parameters: a maximum waiting time of 5 minutes, a cost per kilometer of a bus of 6.99 euros, no cost for taxis per kilometer, no cost for buses per trip, a cost per trip of a taxi of 19.71 euros, a cost per hour of an accompanying person of 15 euros, a loading time of a user into the vehicle of 3 minutes, an unloading time of a user of 1 minute, a number of adapted taxis of 4, a number of nonadapted taxis of 4, a number of buses of

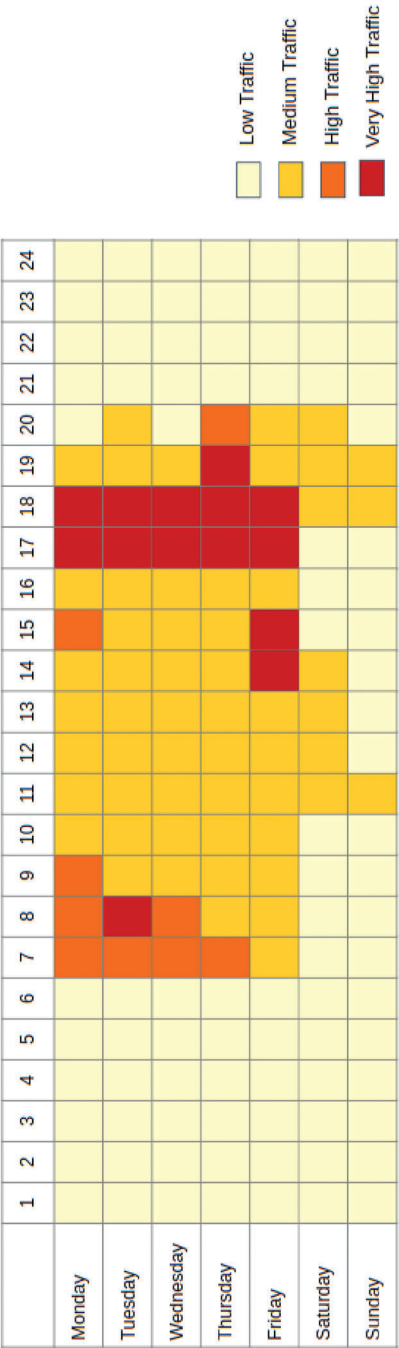


Fig. 3. The figure shows the cluster assignment for each weekday and hour based on the duration of trips between various origin–destination pairs. The cells are color-coded to indicate the cluster to which each weekday hour belongs.

Table 8

Comparison of key metrics between the SMTS and the solutions of our model for each of the five routes (A, B, C, D, E)

| Route                       | Model     | Total distance (km) | Companion cost (€) | Maximum user time (minutes) | Total user time (minutes) | No. of buses | No. of taxis | Average user time (minutes) | Total cost (€) |
|-----------------------------|-----------|---------------------|--------------------|-----------------------------|---------------------------|--------------|--------------|-----------------------------|----------------|
| A                           | SMTS      | 11.84               | 15.00              | 51.11                       | 225.97                    | 1.00         | 0.00         | 32.28                       | 97.78          |
|                             | Our model | 9.28                | 15.00              | 50.00                       | 230.71                    | 1.00         | 0.00         | 32.96                       | 79.86          |
| B                           | SMTS      | 15.57               | 30.00              | 46.11                       | 209.36                    | 1.00         | 0.00         | 26.17                       | 138.80         |
|                             | Our model | 18.44               | 15.00              | 30.65                       | 133.47                    | 1.00         | 1.00         | 16.68                       | 108.34         |
| C                           | SMTS      | 14.97               | 30.00              | 58.48                       | 225.30                    | 1.00         | 0.00         | 37.55                       | 134.65         |
|                             | Our model | 16.02               | 15.00              | 56.00                       | 180.48                    | 1.00         | 1.00         | 30.08                       | 97.64          |
| D                           | SMTS      | 14.44               | 15.00              | 54.17                       | 173.92                    | 1.00         | 0.00         | 34.78                       | 115.91         |
|                             | Our model | 10.06               | 15.00              | 47.04                       | 115.18                    | 1.00         | 0.00         | 23.04                       | 85.29          |
| E                           | SMTS      | 18.58               | 30.00              | 63.79                       | 232.59                    | 1.00         | 0.00         | 33.23                       | 159.88         |
|                             | Our model | 16.85               | 15.00              | 50.29                       | 155.84                    | 2.00         | 1.00         | 22.26                       | 104.64         |
| Relative change average (%) |           | −7.16               | −30.00             | −14.85                      | −24.17                    | 20.00        | na           | −24.17                      | −25.75         |

4, a maximum ride time per vehicle of 480 minutes, and a maximum ride time per passenger of 75 minutes. We have set a limited number of taxis in our model, although in reality, the number is unlimited. This number was chosen based on the number of users per route and has been verified not to affect the results.

These parameter settings were carefully selected with the help and knowledge of the IMPD to optimize the transportation routes and costs in our model while ensuring acceptable limits for users.

#### 6.4.2. Comparison of the SMTS route and our solution

The results of our experiments are summarized in Table 8. The table compares the performance of the current SMTS route with the routes generated by our optimization model. The comparison is based on five routes: A, B, C, D, and E. The table includes the following columns: *Route* shows the route evaluated and whether the results are from the current route of the SMTS or the ones obtained with our model. *Total distance* shows the total distance traveled by the vehicle. *Companion cost* shows the total cost of the accompanying persons. *Maximum user time* shows the maximum amount of time a user spends on a vehicle. *Total user time* shows users' total time on vehicles. *No. of buses* shows the number of buses used. *No. of taxis* shows the number of taxis used. *Average user time* shows users' average time on vehicles. *Total cost* shows the total cost of the vehicle routes. Our predictive model presented in the previous section obtains the duration used for both the SMTS route calculation and our routes. The last row of the table shows the average relative change between the previous results and the new results for each metric. This can be used to assess the overall performance of the new model compared to that of the SMTS routes.

The results of our transportation model show significant improvements over the previous model in several key areas.

First, our model reduces the total vehicle distance traveled by an average of 7.16% across all routes. This indicates that our model is able to find more efficient routing solutions, reducing the overall fuel consumption and carbon emissions of the transportation system.

Table 9

Results of varying pickup time windows and maximum user time in a vehicle

| Time windows (minutes)   | Total distance (km) |       | Maximum user time (minutes) |       | Total user time (minutes) |        | No. of buses |      | No. of taxis |      | Average deviation (minutes) |       | Average user time (minutes) |       | Total cost (€) |        |
|--------------------------|---------------------|-------|-----------------------------|-------|---------------------------|--------|--------------|------|--------------|------|-----------------------------|-------|-----------------------------|-------|----------------|--------|
| Max. user time (minutes) | 45                  | 75    | 45                          | 75    | 45                        | 75     | 45           | 75   | 45           | 75   | 45                          | 75    | 45                          | 75    | 45             | 75     |
| 5                        | 21.63               | 22.15 | 29.35                       | 30.39 | 115.52                    | 118.95 | 2.2          | 2.0  | 1.4          | 1.6  | 3.93                        | 3.60  | 17.40                       | 17.86 | 138.76         | 137.30 |
| 15                       | 16.09               | 15.19 | 36.68                       | 41.01 | 131.60                    | 145.19 | 1.6          | 1.6  | 1.0          | 0.8  | 9.55                        | 9.28  | 19.90                       | 22.11 | 112.73         | 109.37 |
| 30                       | 16.67               | 14.64 | 39.54                       | 44.06 | 133.96                    | 153.33 | 1.4          | 1.4  | 1.0          | 0.6  | 19.16                       | 20.06 | 20.73                       | 23.42 | 106.63         | 105.18 |
| 45                       | 15.42               | 14.13 | 40.27                       | 44.79 | 142.82                    | 152.86 | 1.2          | 1.2  | 0.8          | 0.6  | 26.00                       | 28.88 | 21.87                       | 23.32 | 104.42         | 101.37 |
| 60                       | 15.42               | 14.13 | 39.99                       | 46.80 | 144.66                    | 163.13 | 1.2          | 1.2  | 0.8          | 0.6  | 37.08                       | 35.30 | 22.15                       | 25.00 | 104.42         | 101.37 |
| Overall average          | 17.05               | 16.05 | 37.17                       | 41.41 | 133.71                    | 146.69 | 1.52         | 1.48 | 1.00         | 0.84 | 19.14                       | 19.42 | 20.41                       | 22.34 | 113.39         | 110.92 |

The table shows the total distance (km), the maximum and total user time (minutes), the number of buses and taxis used, the average deviation (minutes) from the desired pickup time, the average user time (minutes) and the total cost (€) for different time windows (minutes), and maximum user times (minutes). The last row presents the calculated average values for the corresponding columns.

Second, our model is able to reduce the average user time in the vehicle by an average of 24.17% (7.80 minutes) across all routes, saving the user 26 hours in the vehicle at the end of the year (during the weekdays from September to June). This indicates that our model provides faster and more efficient transportation for users, reducing their overall travel time and improving their experience.

Third, our model is able to reduce the total cost by an average of 25%. This implies a weekly reduction cost of €856.25, implying a reduction to the annual cost of approximately €34,250.

Overall, we found that the optimal solutions generated by our model outperform the current SMTS routes in terms of reducing vehicle distance, total costs, and user travel time. These improvements can be attributed to the ability of our model to use multiple buses and taxis, whereas the SMTS uses only one bus for each route. This suggests that our model provides more efficient and effective transportation services, potentially leading to cost savings and improved user satisfaction.

#### 6.4.3. Impact of varying pickup time windows and maximum user ride time

In this section, we evaluate the effects of varying the pickup time window parameters and the maximum time of a user in the vehicle. By doing so, we can provide guidelines to public transportation operators on which parameters to use and how the parameter values affect their services. Therefore, the goal is not to find the best parameter value but rather to show how the change in the parameters affects the service and in which ways. We examine the effects on the total cost and other indicators such as the total travel distance of the vehicles, the maximum time a user spends on a vehicle, and the number of vehicles needed.

Table 9 shows the results obtained by varying the values of the pickup time windows and the maximum time spent by a user in a vehicle. These values are the average of the calculated routes. In addition to the attributes seen in the previous table, we have included a new measure (*average*



*deviation*) to see the average deviation between the desired pickup time of the users and the pickup time selected by the route given the time window.

Based on the values in Table 9, we can see how the total distance traveled by vehicles and the total cost decrease as the time window increases. This is likely because having more time allows for greater flexibility in vehicle routing, and since the objective function of our model is to minimize costs, this is prioritized. The number of buses and taxis also decreases as the time window increases, given the flexibility in vehicle routing. By solving instances of larger requests, the ratio of vehicles per user continues to decrease because we can better utilize the capacity of the vehicles and minimize the number of vehicles needed to fulfill the requests. This can ultimately result in lower costs and increased profitability.

The average deviation from the planned routes increases as the time window increases, indicating that as we have a larger margin of the time windows, the assigned pickup time can deviate from the desired pickup time. When comparing the maximum time a user can spend inside a vehicle, we observe that reducing the maximum time from 75 to 45 minutes resulted in a minimum increase in the number of taxis needed, as well as the total distance and cost. However, this decrease in maximum time also reduces the total user time and average user time spent in the vehicle, increasing the quality of the service for the users.

While Table 9 offers insight into various performance metrics, it is important to consider the runtime of the algorithms. Here, we present an analysis of the average runtime observed over the five routes evaluated, revealing a clear trend of increasing runtime with the complexity of the instances.

For the smaller instances with a time window of just 5 minutes, the runtimes were relatively swift at 2.69 and 3.33 seconds for maximum user times in a vehicle of 45 and 75 minutes, respectively. However, as we expanded the time window to 15 minutes, these values jumped to 24.99 seconds and 31.07 seconds for the same maximum user times. A more pronounced surge in runtime was noted when the time window was set at 30 minutes, with times of 97.98 and 113.22 seconds. The increase continued for larger time windows of 45 and 60 minutes, which had maximum runtimes of 108.90 and 244.99 seconds, respectively, for the two different maximum user times.

It is evident that as the complexity, encapsulated by the time window and maximum user time, rises, so does the computational demand of our algorithm. While the algorithm handled the smaller instances competently, future work should be focused on optimizing the algorithm further to manage larger instances efficiently.

## 7. Conclusions

In this paper, we introduced a new variant of the DARP, the rHDARP, which considers the daily door-to-door transportation of people with reduced mobility in the city of Barcelona. We proposed a new MILP to solve the rHDARP, which includes a fleet of buses and taxis, multiple depots, time windows at pickup and delivery locations, maximum ride and waiting times, and the possibility of an accompanying person when needed.

We demonstrated the effectiveness of our proposed solution by conducting a case study of the SMTS in Barcelona. Our solution significantly reduced the distance and the costs of the regular routes of SMTS while also improving the quality of service for the users.

One of the key challenges in the rHDARP was the variation in trip duration based on the time of day and day of the week. To address this challenge, we developed a predictive model that utilized machine learning techniques to accurately estimate trip durations at various times of the day and on different days of the week.

Our analysis of the influence of the time window and maximum ride time parameters on the cost and quality of service of SMTS provides insights for public transportation operators to make more informed decisions.

In future work, applying our solution to other transportation services and incorporating other relevant factors, such as traffic congestion and weather conditions or electric vehicles, can be explored. We also plan to conduct a more thorough study involving various metaheuristic algorithms to solve the problem with larger instances for our proposed rHDARP.

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